

Java programs

1. Reverse a Number

A banking application stores customer transaction IDs as integers. For security verification, the system needs to reverse the digits of the transaction ID before processing. Write a Java program to read an integer from the user and display its reversed form.

Example:

Input: 1234

Output: 4321

2. Check Palindrome Number

An e-commerce company generates product serial numbers. Some serial numbers are considered special if they read the same forward and backward. Write a Java program to check whether the given serial number is a palindrome.

Example:

Input: 121

Output: Palindrome

3. Find Largest of Three Numbers

A warehouse management system records the quantity of three different products received on a particular day. The manager wants to identify which product has the highest stock. Write a Java program to find the largest among three given numbers.

Example:

Input: 10 20 5

Output: 20

4. Count Vowels in a String

A content analysis tool needs to determine how many vowels are present in a given text for readability analysis. Write a Java program that accepts a string and counts the total number of vowels present.

Example:

Input: amazon

Output: 3

5. Remove Duplicates from a String

A text-cleaning application removes repeated characters while keeping the order of first appearance unchanged. Write a Java program to remove duplicate characters from a given string.

Example:

Input: programming

Output: progamin

6. Two Sum Problem

An online shopping application stores product prices in an array. A customer wants to buy exactly two products whose total price matches a given budget. Write a Java program to find and print the two product prices.

Example:

Input: [2,7,11,15], target=9

Output: 2 7

7. Find Missing Number

A school maintains roll numbers from 1 to N for students. Due to a data entry error, one roll number is missing. Write a Java program to identify the missing roll number.

Example:

Input: [1,2,4,5]

Output: 3

8. Longest Substring Without Repeating Characters

A communication system analyzes messages and wants to find the longest sequence of characters without repetition. Write a Java program to determine the length of the longest substring without repeating characters.

Example:

Input: abcabcbb

Output: 3

Explanation:

"abc" is the longest substring.

9. Merge Two Sorted Arrays

A logistics company receives two sorted delivery schedules from different branches and wants to combine them into a single sorted schedule. Write a Java program to merge the two sorted arrays.

Example:

Input:

[1,3,5]

[2,4,6]

Output:

[1,2,3,4,5,6]

10. String Compression

A data storage system compresses repeated characters to save memory. Each repeated character is replaced by the character followed by its count. Write a Java program to compress the given string.

Example:

Input: aaabbc

Output: a3b2c1